

Goal. Find approximations to definite integrals

1. Fill in the table below with the data you will need to approximate the definite integral of $f(x) = e^{-x^2}$ from 1 to 3 with $n = 6$.

k	x_k	$f(x_k)$
0		
1		
2		
3		
4		
5		
6		

Also, what does Δx equal?

2. Use the data above to find L_6 , the left endpoint approximation.
3. Use the data above to find R_6 , the right endpoint approximation.
4. Use your work in Problems 2 and 3 to find T_6 , the trapezoidal rule approximation.
5. Fill in another data table to find M_6 , the midpoint rule approximation.

k	$\bar{x}_k$	$f(\bar{x}_k)$
1		
2		
3		
4		
5		
6		

6. Use your previous work to find S_{12} , Simpson's rule approximation.

It is important to know whether these approximations are accurate. Fortunately, there is a known **error bound** for each of the approximation rules that tells us how close our approximation must be to the true value of the integral. The **error** is defined as the difference between the value of the integral and the approximation. For example, the error in the trapezoidal rule approximation, which we write as $\text{Error}(T_n)$, is $\left| \int_a^b f(x) \, dx - T_n \right|$. A bound on the error is given as: $\text{Error}(T_n) \leq \frac{K_2(b-a)^3}{12n^2}$ where $|f''(x)| \leq K_2$ for $a \leq x \leq b$.

7. Use the above error bound given above to determine how close our T_6 approximation from Problem 4 is guaranteed to be to the actual value of the integral. Hint: To find K_2 , find the maximum value of the absolute value of the second derivative of the function on the given interval. You may use Desmos or another online plotting system to help you.

You can (BUT DO NOT NEED TO) do the same for M_6 from Problem 5 using the error bound for the midpoint rule below.

$$\left| \int_a^b f(x) \, dx - M_n \right| = \text{Error}(M_n) \leq \frac{K_2(b-a)^3}{24n^2} \quad \text{where } |f''(x)| \leq K_2 \text{ for } a \leq x \leq b.$$

8. How big would n have to be so that M_n is within .0001 of the actual value?

You could also use the error bound below to determine how close S_{12} is guaranteed to be to the actual value of the integral (BUT DO NOT NEED TO). (Note that $|\frac{d^4}{dx^4}e^{-x^2}| \leq 8$ for $1 \leq x \leq 3$.)

$$\left| \int_a^b f(x) dx - S_n \right| = \text{Error}(S_n) \leq \frac{K_4(b-a)^5}{180n^4} \quad \text{where } |f^{(4)}(x)| \leq K_4 \text{ for } a \leq x \leq b.$$

9. How big would n have to be so that the trapezoidal rule approximation T_n for the integral $\int_1^3 e^{-x^2} dx$ is within .0001 of the actual value?

(Hint: Substitute the values of K_2 , a , and b you used in Problem 8 into the inequality $\frac{K_2(b-a)^3}{12n^2} \leq .0001$ and then solve the inequality for n .)

Spend any remaining class time working on your project.